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Transaction Fee Mechanisms and User Welfare in Permissionless Blockchains

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Abstract

A transaction fee is fundamental in maintaining the stability and decentralization of a permissionless blockchain, the infrastructure underlying major cryptocurrency networks such as Ethereum and Bitcoin. This fee, set and paid by users when initiating each transaction, aggregates to over $24 million daily on Ethereum, imposing a nontrivial barrier to wider adoption. What is worse is that it is disproportionately borne by small users. Like Bitcoin, Ethereum started with a name-your-own-price style fee mechanism. Under this Legacy mechanism, we demonstrate that, surprisingly, only three-quarters of the transaction fee is necessary for users to attain the service quality they receive; the remaining one-quarter thus constitutes welfare loss. In August 2021, Ethereum implemented a fee mechanism change. Together, the Legacy methods and the new scheme, which we call Cap-Tip, describe the two most widely used fee mechanisms in all major permissionless blockchains. We explain the innovations embedded in the Cap-Tip mechanism through the lens of information quality and task decomposition. Collecting over 123 million transactions covering the change period, we estimate the causal impact of Cap-Tip on welfare improvement using a doubly robust difference-in-difference estimator that accommodates heterogeneity across timing of and exposure after adoption. The result is an astounding 80% reduction in welfare loss, achieved by users only 6 days after adoption. Interestingly, we also find that adoption benefits small users more than large ones, alleviating fee disparity and improving fairness. We theorize this difference through user experience and the overconfidence bias. Our study offers a novel operational perspective to unpack how the arguably most significant fee mechanism change in the blockchain space affects user welfare.

Keywords

Blockchain, User Welfare, Priority Queues, Congested Service Fee, Staggered DiD

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1 Introduction

Blockchain technology improves the transparency and verifiability of transactions (Babich and Hilary, 2020; Mithas et al., 2022a) by recording them on distributed ledgers. It holds promising application values in business domains such as crowdsourcing (Tsoukalas and Falk, 2020), counterfeit detection (Pun et al., 2021; Shen et al., 2022), and peer-to-peer lending (Chung et al., 2023). Without centralized control, *permissionless* blockchains are developed and maintained by miners under the proof-of-work (PoW) or validators under the proof-of-stake (PoS) consensus mechanisms.[1] They generate new blocks, pack transactions into these blocks, and append them on top of the existing block to form a blockchain.[2] Advocates of the permissionless blockchain often cite the accessibility it extends to unbanked or underbanked populations to carry out financial activities (Andreasson, 2022). This morally prominent goal, however, needs to be viewed against the capacity constraints of the cryptocurrency systems. That is, to ensure the decentralization and security of a permissionless blockchain system, it is inevitable that it will experience scalability limits, which is commonly known as the blockchain trilemma.[3] As of early 2023, the Bitcoin

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blockchain processed approximately 3 transactions per second, while Ethereum processed 13 transactions per second,[4] both of which are substantially lower than the transaction throughput of established financial systems such as VISA, capable of handling approximately 5,000 transactions per second (Malik et al., 2022)—equivalent to about 1,600 times Bitcoin's throughput and 400 times that of Ethereum. Therefore, it is not difficult to imagine that demand frequently outweighs capacity in leading permissionless blockchain systems, making accessibility highly dependent on priority rules.

Departing from the widely studied first-come-first-serve (FCFS) priority rule, permissionless blockchains allow transaction senders to name their own transaction fee, an add-on fee paid to miners, and assign priority based on this user-determined fee. This departure from FCFS is, to a certain degree, analogous to emergency departments prioritizing patients according to a severity index (Bolandifar et al., 2019) and amusement parks offering customers access to a higher priority queue for a fee (Kostami and Ward, 2009). Blockchain is, however, different from these non-FCFS systems in the absence of a centralized platform operator who directly assigns priority or offers a menu of service speed options. Although name-your-own-price (NYOP) approach to setting transaction fees grants blockchain users the maximum freedom in controlling their service speed, Basu et al. (2023) and Shang et al. (2023) argue that estimating a required transaction fee that helps achieve a desired speed is difficult for the average user, as the actual service speed is also contingent on other users' fees.

Two user welfare issues emerge as a result of this fee-estimation difficulty. First, given that pending transactions are selected and packed in batches into a newly created block by its miner, any user-named fees above the minimum entry fee (i.e., lowest fee in the block) increase the transaction cost to the user yet do not improve the service quality. Thus, this amount of overpayment is effectively a *user-side welfare loss* metric that can be directly measured using publicly available data. Notably, such overpayment can occur irrespective of the transaction completion speed. The second, and perhaps more worrisome, aspect is the disproportionate effect that this challenge has on users of different sizes. Large senders have the resources to develop sophisticated machine learning algorithms to estimate the desired transaction fee more accurately and thus reduce their welfare loss,[5] whereas small senders, including the unbanked or underbanked population, must rely on off-the-shelf simple fee recommendation tools provided free of charge by crypto wallets such as MetaMask. This disparity contributes to an unfair transaction fee distribution across users, as shown in Figure 1. Specifically, we sort Ethereum user addresses by the total transaction value from low to high, presenting the cumulative transaction value[6] on the x-axis and the cumulative transaction fee across these addresses on the y-axis. Ideally, users sending a certain fraction of transaction value should bear a corresponding proportion of the transaction fee, i.e., the dashed diagonal line. In

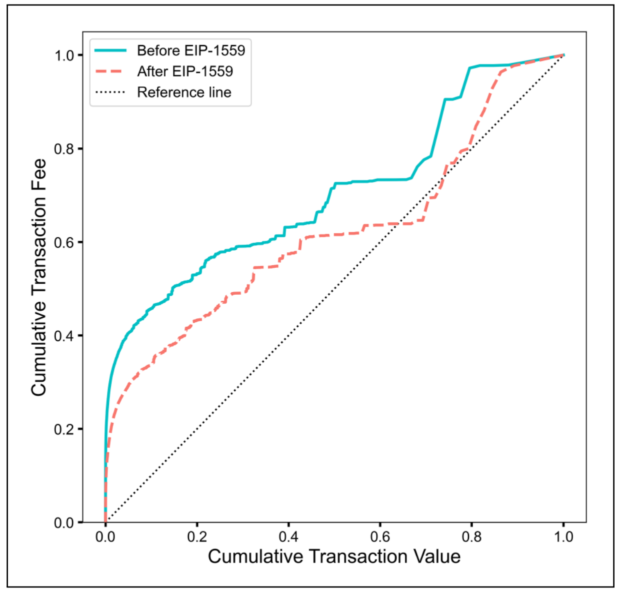

Figure 1. Cumulative transaction fee versus cumulative transaction value.
Notes. This plot illustrates the relationship between cumulative transaction fee (*y*-axis) and cumulative transaction value (*x*-axis, in USD). The green solid and red dashed lines represent this relationship before and after the implementation of EIP-1559, respectively. The black dotted diagonal line represents a socially just distribution of transaction value and fee, where users sending higher value transactions are paying proportionally higher fees.

reality (green solid line in Figure 1), addresses making up the first 2% (10%) of the cumulative transaction value have paid 34% (46%) of all transaction fees, severely deviating from an ideal fee distribution.

The transaction fee mechanism (TFM) discussed above (abbr. Legacy TFM hereafter) was augmented by a new TFM (abbr. Cap-Tip TFM hereafter) on the Ethereum network on August 5, 2021 through the Ethereum Improvement Proposal 1559 (abbr. EIP-1559 hereafter).[7] Figure 2 summarizes the key elements of the Legacy and the Cap-Tip TFMs from a user's perspective. Under Legacy TFM, a user such as User A1 in the figure sets a "Per-gas Fee" when she initiates the transaction. The total transaction fee is then calculated by multiplying this Per-gas Fee with the amount of gas[8] consumed. Once initiated, the transaction enters a pending pool containing all transactions initiated but not yet included in a block. Miners constantly scan the pending pool to "discover" new transactions and rank all pending transactions by the Per-gas Fee. To maximize revenue from transaction fees, the successful miner—the first to solve a trial-and-error puzzle and broadcast her solution to other nodes in the Ethereum network—packs top-ranked transactions into the new block. This miner earns revenue from two sources: (1) transaction fees from senders and (2) minting rewards from the system. Note that each block has a capacity limit, known as the "block size," and the total size of all selected transactions cannot exceed this

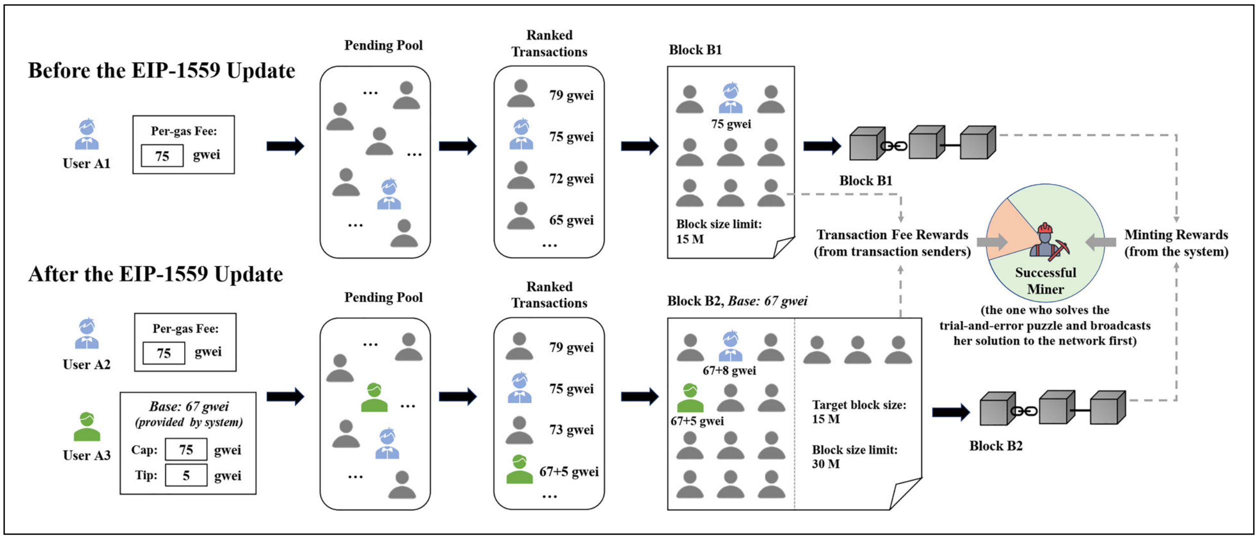

Figure 2. Confirmation process of transactions before (top) and after (bottom) the EIP-1559.
Notes. The Legacy TFM is available both before and after EIP-1559, while the Cap-Tip TFM is available only after EIP-1559. Under Legacy, users specify a "Per-gas Fee," based on which the miners prioritize transactions to be packed into a block. In contrast, Cap-Tip requires users to specify two parameters—Cap and Tip—and pay minBase + Tip, Cap, where Base serves as the minimum entry price into the next block and adjusts dynamically to contemporaneous user demand. Miners prioritize transactions in a similar fashion, yet receive only the fee payment above Base as revenue.

limit. Before EIP-1559, this limit was fixed at 15 million gas. Once verified by more than half of the miners, this block is chained to the existing blocks to form the updated blockchain. This process embodies characteristics of both a priority-based queueing system and a first-price auction. Bitcoin shares a similar transaction confirmation process (Shang et al., 2023: 108).

EIP-1559 enacts three changes. First, the variable-size block is designed to mitigate transient congestion by (partially) accommodating demand surges through a larger block size. Second, a system-provided minimum fee per gas to enter the next block (i.e., the Base) is calculated from a preset formula: $Base_{n+1} = Base_n * \left(1 + \lambda * \frac{Actual\ Block\ Size_n - Target\ Block\ Size}{Target\ Block\ Size}\right)$, where n denotes block number and λ is a predefined parameter that controls the scale of updating. Thus, Base for the future block adjusts upward (downward) when more (fewer) transactions are included in the previous block, which to some extent indicates the system is contemporaneously more (less) congested. Base alleviates the fee-setting difficulty by revealing a reference range: fees below Base are currently not accepted, while fees considerably higher than Base are likely unnecessary.[9] Third, EIP-1559 decomposes the bid parameter of Legacy TFM, "Per-gas Fee," into two distinct decision components: Tip and Cap. Tip reflects the extra fee one would like to offer on top of Base, and Tip + Base works in a first-price auction manner, functionally similar to the user-defined Per-gas Fee in the Legacy TFM. Meanwhile, Cap serves as a safeguard for users, such that total fee per gas equals to minCap, Base + Tip. In other words, if one's transaction is not included in the next block and the Base increases subsequently, Cap can prevent the actual payment from rising above the sender's willingness to pay. It is worth noting that adoption of the Cap-Tip TFM was gradual, reaching about 39% on October 16, 2021. Such staggered adoption means the co-existence of Legacy and Cap-Tip TFMs post-EIP-1559 (i.e., Users A2 and A3 in Figure 2).

While the Cap-Tip TFM offers an obvious information provision advantage over the Legacy TFM via the addition of Base, whether the transition from the single fee-setting decision in Legacy to the dual-decision in Cap-Tip, which increases decision complexity yet enforces task decomposition, is beneficial for an average user remains an open behavioral question. In addition, whether the TFM transition can benefit small users more and hence close the transaction fee disparity is a priori unknown. Thus, we focus our theoretical analysis on these interesting yet uncertain aspects of EIP-1559. Empirically, this transition offers us the only opportunity to compare two TFMs used by mainstream permissionless blockchains. As a result, quantifying the magnitude of overpayment reduction in transaction fees brought forward by the Cap-Tip TFM is of paramount practical relevance. To this end, we propose a novel measure of overpayment, which can be directly computed from archival Ethereum data and is generalizable to other permissionless blockchains. We center our analysis around this measure to offer insights into user-side welfare improvement and answer the socially pressing question of who benefits more from EIP-1559. Encouragingly, Figure 1 shows the actual fee distribution (red dashed curve) is closer to the ideal (dotted diagonal line) in the post-EIP-1559 era.

Our raw data covers all Ethereum transactions between July 6 and October 16, 2021. Since the not-yet-updated users naturally form a clean control group, we implemented a state-of-the-art difference-in-difference approach with staggered adoption (Callaway and Sant'Anna, 2021), as recommended by Mithas et al. (2022b). Specifically, we estimated the reduction in user-side welfare loss attributed to the EIP-1559 transition for each combination of adoption timing (i.e., a group of users adopting on the same date) and days after adoption. These highly granular, group-time level treatment effects were then aggregated to offer a rich set of insights. First, we show that the Cap-Tip TFM reduces user-side welfare loss by 80%. Further attributing this effect to Base and Cap, the two main modifications introduced by the new TFM, reveals that the former captures most of the overpayment reduction. Second, user learning of Cap-Tip is fast: it takes only 6 days for the effect to saturate. This learning is mostly on setting Tip. Third, small users take fewer practice rounds for Cap-Tip's effect to rise to its full force. This is because small users exercise more discipline in adhering to the design principles of the Cap-Tip TFM than large users do. Last, despite a common concern raised in the literature that Cap-Tip's performance over Legacy might not sustain during episodes of high congestion, our results show the contrary: Cap-Tip exhibits a stronger benefit on more congested days.

2 Literature Review and Contribution

Our study draws upon and contributes to three streams of literature: (1) blockchain transaction fees, (2) data transparency in blockchain-based systems, and (3) user learning and overconfidence.

Ilk et al. (2021) conceptualize cryptocurrency platforms (e.g., Bitcoin) as data-space markets, investigating how miners' supply and users' demand behaviors shape the equilibrium of transaction fees. Building on this, Shang et al. (2023) quantify the relationship between transaction fee and speed in the Bitcoin market, developing a recommendation system to optimize fees based on target service levels. Absent in their investigations is the attention to user welfare (under the legacy TFM). To this end, Basu et al. (2023) propose an alternative TFM based on uniform-price auctions, demonstrating its theoretical potential to reduce users' transaction costs and decrease miners' income variance. Nevertheless, this proposed TFM, to our knowledge, has not been adopted by any major permissionless blockchain systems. Thus, its real-world performance remains unknown. In contrast, the focal event of our study, EIP-1559, triggers a transition from the Legacy TFM to the Cap-Tip TFM, two widely used fee mechanisms by mainstream blockchains. Roughgarden (2020) and Ferreira et al. (2021) provide theoretical analyses on the latter, showing its potential to reduce fee volatility and transaction delays for rational users. Importantly, the former further argued that under equilibrium conditions, users' decisions on Cap should equal their willingness to pay, similar to the equilibrium Per-gas Fee amount under Legacy TFM, and that Tip should be set to a negligible amount equal to the miner's marginal cost of processing one more unit of gas (Roughgarden, 2020, Section 6.3). Conversely, the latter cautioned that the Cap-Tip TFM still lets users compete in a way similar to a first-price auction and, hence, may not perform better than Legacy TFM under high system congestion. Reijsbergen et al. (2021) also suggest that Cap-Tip does not perform well during demand bursts and thus propose an improved updating rule for Base. In addition, Liu et al. (2022) report preliminary empirical evidence of user experience improvements post-EIP-1559. While their analysis shows a reduction of intra-block variance of gas price, there is a lack of more granular analysis of user-side welfare at the address or transaction level. Therefore, our study provides the first granular and systematic empirical analysis on user-side welfare loss under the two popular TFMs. We also decompose the welfare improvements of Cap-Tip into its individual components, offering clear theoretical insights for future fee mechanism designs.

The second stream of literature explores the implications of blockchain-enabled transparency and traceability. Permissionless blockchains are often touted with radical transparency. Cui et al. (2023) utilize a game-theoretic model to demonstrate that blockchain traceability enhances product quality and corporate profits by mitigating dual moral hazard. Similarly, Dong et al. (2023) find that blockchain traceability benefits all food supply chain members by reducing waste of uncontaminated products. Furthermore, several studies have highlighted the potential of blockchain transparency in counterfeit prevention. For instance, Pun et al. (2021) suggest that manufacturers can use blockchain technology to verify product authenticity in markets plagued by counterfeiting. Shen et al. (2022) demonstrate that blockchain technology, with its decentralization and transparency characteristics, is an effective solution for combating counterfeit issues in the supply chain. In contrast, Sriraman et al. (2023) and Zhan et al. (2025) use failed blockchain application cases to demonstrate that data transparency alone does not solve real-world problems. Similarly, Zhao et al. (2022) argue that the transparent yet fragmented on-chain raw data does not directly offer sufficient help to voters for casting votes in a decentralized autonomous organization (DAO). Our study contributes to the debate in this literature stream by demonstrating that even with complete data transparency, permissionless blockchain users can still considerably benefit from the provision of compiled information based exactly on this transparent data.

The third literature stream relates to experience and overconfidence. Soll et al. (2022) find that alignment between individuals' initial judgments and advisors' recommendations can induce overconfidence, impairing their subsequent performance. Similarly, Logg et al. (2019) have participants complete a series of prediction tasks and show that experienced professionals are less likely to rely on algorithmic recommendations, reducing their estimation accuracy. Our study contributes to this literature stream by unveiling an intriguing phenomenon: experienced large users benefit less than smaller users from adopting the Cap-Tip TFM due to their overconfidence. Interestingly, such behavioral difference across users also produces an unintended but positive outcome for the system as a whole: the transaction fee disparity between large and small users is mitigated to a certain extent after EIP-1559. This also adds to the broader literature on fairness concerns in operational design (Bertsimas et al., 2012). While many resource allocation problems involve a trade-off between fairness and efficiency (Papalexopoulos et al., 2023), the Cap-Tip TFM improves fairness at no loss of efficiency.

3 Hypotheses Development

From the user's perspective, changes enacted by EIP-1559 can be viewed as first providing the Base as support information, as setting Tip is comparable to setting the Legacy Per-gas Fee with Base and then adding Cap to the decision set. We theorize the benefits of these two elements in Sections 3.1 and 3.2. Note that the dynamic block size is captured by Base and hence does not add further information to fee-setting. Section 3.3 explores how these benefits vary across small and large users.

3.1 How Does Base Improve Information Quality?

Blockchain technology is recognized for its ability to make on-chain data transparent. However, data transparency does not directly translate into easiness of on-chain decision-making. Under the Legacy TFM, users rely entirely on themselves to collect information needed to name the transaction fee. In addition, as discussed in Shang et al. (2023: 107–108), the exact block-admission priority of a transaction depends not only on pending transactions but also on future transactions submitted afterward. For a sophisticated user, higher-fee transactions in the former group can be calculated exactly. In contrast, those in the latter group, known as effective arrival rate in the queueing literature (Guo and Zipkin, 2007), need to be forecasted. Unfortunately, for a casual user with limited resources, this is too daunting a task. This partly contributes to the fee payment disparity between small and large addresses.

Instead of relying on self-constructed congestion signals in the Legacy TFM, users are offered a system-provided information cue (i.e., Base) after the EIP-1559 update. This directly accessible congestion signal greatly simplifies the fee-setting decision. We explain this through the lens of information quality theory, which states that the effectiveness of information can be assessed from four dimensions: completeness, accuracy, format, and currency (Xu et al., 2013). These dimensions describe how users perceive the information provided by the system, including how fully it supports decision-making (completeness), how correct it is (accuracy), how effectively it is presented (format), and how up-to-date it is (currency). Interestingly, information format remained largely unchanged before and after EIP-1559, with most users still accessing transaction records via information aggregators and initiating transactions through wallets.[10] In addition, information accuracy is guaranteed by design in permissionless blockchains through the use of cryptography. Thus, we emphasize how Base enhances information completeness and currency in the following. We draw analogies from literature streams on delay announcements in queues and reserve price in auctions.

A strand of empirical queueing literature investigates how service providers can "announce" waiting-related information to improve system performance (Yu et al., 2021) and increase customer surplus (Dong et al., 2019). Guo and Zipkin (2007: 962) refer to system occupancy as partial information (partial completeness) and waiting time as full information (high completeness) and find that full information usually creates better system performance than partial information. Wang and Hu (2020) demonstrate that real-time queue-length information (high currency and partial completeness) offers customers insights into congestion levels and consistently leads to higher social welfare compared with the no announcement scenario. The public reserve price in an English auction provides another analogy for the information provision role of Base. Set by the auctioneer (i.e., seller), the reserve price communicates the minimum price required for a bid to win the auction (partial completeness). Trautmann and Van de Kuilen (2015) argue that the revealed reserve price serves as a reference point for bidders. Han et al. (2021) show empirically that when there is no public reserve price, the number of competing bidders increases, resulting in a higher final price.

Thus, similar to delay announcement in queues and reserve price in auctions, Base conveys the entry price for the next block and establishes a reference point for users deciding their total transaction fee (i.e., Base + Tip), enhancing information completeness. Furthermore, Base is updated by system in real-time (around every 13 s) according to the actual usage of the most recent block. As the data collection and processing speed of regular users is likely to be slower, Base also enhances information currency. Given the information provision role of the Base fee, we propose the following hypothesis:

Hypothesis 1: Provision of the Base fee information reduces permissionless blockchain users' transaction fee overpayment.

3.2 How Do Cap and Tip Enforce Task Decomposition?

Prior literature on online decision-making highlights task's structure (well-structured vs. ill-structured) and solver's cognitive strategy (decompositional vs. holistic) as two main drivers of task performance; see Browne et al. (2007) for a review. The fee-setting task for blockchain users is well-defined: achieving the desirable transaction confirmation speed with the minimal transaction fee possible. The nature of this problem and hence the task structure remain unchanged before and after EIP-1559. Thus, in the following, we focus on how EIP-1559's modification of a single Per-gas Fee (one decision variable) into a combination of Tip and Cap (two decision variables) alters user's cognitive strategy.

When decision-makers are presented with a large amount of information to sift through (i.e., high information processing load), they tend to avoid reductionist reasoning and instead resort to a holistic solution strategy (Browne et al., 2007: 93). Lee and Siemsen (2017) demonstrate a similar preference for the holistic strategy when the decision environment is uncertain. The fee-setting task exhibits both features: high information load from both recent on-chain and pending transactions and high uncertainty as the same fee might lead to varying service speed depending on contemporaneous system congestion. Thus, it is conceivable that users are naturally resistant to engaging in beneficial task decomposition.

With the implementation of EIP-1559, the system enforces task decomposition. That is, users now are forced to think about Cap and Tip separately. As shown in the theoretical analysis of Roughgarden (2020, Section 6.3), this decomposition makes the optimal solution obvious: setting Cap to willingness to pay and setting Tip to a value (usually small) that can compensate miners' marginal cost. Although users might still deviate from the above optimal solution, connecting Cap with one's internal willingness to pay and Tip with the external competition certainly moves the paid fee, minCap, Base + Tip, in an overpayment-reducing direction. This benefit of system-enforced task decomposition is in line with Lee and Siemsen (2017): once experimental subjects are instructed to decompose the inventory ordering task into three subtasks (point forecast, uncertainty estimate, and service-level decision), their performance improves. Thus, we propose:

Hypothesis 2: Changing from a single-variable Per-gas Fee decision to a dual-variable Cap-Tip decision reduces permissionless blockchain users' transaction fee overpayment.

3.3 Small Versus Large Users

Unlike the Legacy TFM, where naming the Per-gas Fee relies entirely on one's collection and digestion of system congestion signals, the Cap-Tip TFM considerably reduces the need for information search and the reliance on experience and judgment. Notwithstanding EIP-1559's promised benefits, the degree to which a user can appropriate these benefits depends on her adherence to the design principles of EIP-1559. We categorize transaction senders into large versus small users based on their past (average) transaction value and argue that large users are likely to enjoy less benefit from the TFM transition due to overconfidence. Prior literature has drawn an analogy between crypto transactions and speculative trading (e.g., Wei and Dukes, 2020), where large users, who possess more economic (money) and noneconomic (knowledge and skill) resources, are likely to gain extensive trading experience, particularly successful experience.

Overconfidence, which describes the bias of having overly positive expectations about one's abilities and judgments (Johnson and Fowler, 2011), causes the decision-maker to discount advice from both humans (Ecken and Pibernik, 2015) and algorithms (Logg et al., 2019). Such bias is common among people who have accumulated considerable experience in a certain field. For example, Gaba et al. (2023) find that experienced managers tend to be overconfident in their abilities, leading them to overlook the need to adjust strategies when receiving negative performance feedback. Similar effects of overconfidence are also reported in financial investments (Park et al., 2013) and CEOs' acquisition activities (Billett and Qian, 2008). Thus, the paradigm changes of EIP-1559 might work against large users, as their successful trading experiences from the pre-EIP-1559 era may lead to overconfidence in their abilities and strategies, leading them to deviate from the simple fee-setting heuristic of the Cap-Tip TFM.

Hypothesis 3a: The Legacy to Cap-Tip TFM change benefits small users more than large users.

Although users consider both external competition and internal willingness to pay when setting the Per-gas Fee under Legacy TFM, we believe it is the former that primarily drives their fee decision. This is echoed by existing studies on Bitcoin transaction fees (Basu et al., 2023; Shang et al., 2023). Therefore, when bidding with the new TFM, users may view "setting Tip + Base given Base" as a post-EIP-1559 parallel to "setting Legacy Per-gas Fee given Base." In other words, users may associate their Tip decisions with their past experience in setting the Legacy Per-gas Fee. Due to the complexity and uncertainty of competitive bidding, large users with sufficient resources are likely to develop their personal strategies in setting the Legacy Per-gas Fee over time, which may lead to overconfidence and hence resistance to the necessary adjustments in setting Tip. Cap, on the other hand, is a newly introduced decision variable with a distinct functionality: limiting the paid fee in a safe range. As a result, we believe the decision bias of large users will primarily manifest in setting Tip. We formalize the above discussion as follows:

Hypothesis 3b: The differential benefit between small and large users is attributed to their way of setting Tip.

4 Data and Identification

4.1 Data Collection

Our main data source is the Ethereum blockchain. We synchronize the public on-chain Ethereum data into our server clusters, which include information at both transaction and block level. Although EIP-1559 was rolled out in the London Hard Fork on August 5, 2021—the earliest time one can use the Cap-Tip TFM in principle—actual adoption happened in a rather gradual fashion. We set our observation window from July 6, 2021 (30 days before London Hard Fork) to October 16, 2021 (30 days after the adoption time of September 16, 2021), yielding a total of 661,554 blocks and 123 million transactions. Empirically, the 6 weeks between August 5, 2021 and September 16, 2021 provide treatment effect heterogeneity on the adoption-timing dimension, while the 30 days after each adoption date provide heterogeneity on the post-adoption learning dimension. We complement our main data source by (1) the label word cloud provided by Etherscan and (2) the CoinMarketCap APIs. The former helps identify and exclude institutional addresses, which are not normally concerned about transaction fees. The latter provides access to day-level prices of ETH and all crypto tokens deployed on Ethereum.

4.2 Data Processing

Processing and cleaning of raw data require fastidious attention to detail. In the following, we concisely document this process while relegating certain details to the Online Appendix.

4.2.1 Measurement of User-Side Welfare and Removal of Private Transactions. In analytical models, user welfare[11] (a.k.a. user surplus) is conveniently expressed as the difference between product valuation and product price, i.e., $v-p$ (Bapna et al., 2008). The main difficulty of operationalizing this measure is the unobservability of v. Exploiting blockchain's unique "admission by block" feature, we propose a user-side welfare loss measure that circumvents the unobservability of v and implements directly on the transaction-level data. Specifically, after sorting the fee per gas of transactions in a block, the lowest fee essentially acts as a de facto minimum price that a sender (of any transaction in the block) would have to pay to receive the same service speed (same v). The difference between this minimum and the actual fee per gas paid by the sender (i.e., the overpayment) is her welfare loss: $v - p_{min} - (v - p_{actual}) = p_{actual} - p_{min}$. The rationale behind this measure is consistent with the price-induced welfare loss often found in analytical models (e.g., Fang et al., 2021). However, how to accurately capture this minimum fee from a block requires extra caution due to the frequent presence of private transactions, which are either the miner's own or negotiated with a third party off-chain (Piet et al., 2022). They need to be removed to establish a meaningful minimum entry fee, as they often pay zero or considerably lower fees. The nuances in the removal process are explained in Appendix B. Overpayment, the primary dependent variable in the following analyses, measures the difference between a transaction's paid fee per gas and the block's non-private minimum entry fee. A total of 117,403,702 transactions from 14,393,803 addresses are retained.

4.2.2 Construction of Treatment and Control Groups. Under staggered adoption, the control group can be constructed from either not-yet-adopted units or from never-adopted units. The former has the advantage of being more similar to the treated units, while the latter might offer a larger control sample size (Callaway and Sant'Anna, 2021: 204–205). We use the former for our main analysis, while the latter is considered in a robustness check. We first screen the addresses in terms of adoption time, adoption pattern, and transaction activeness and then filter out institutional addresses. Institutional addresses (e.g., those controlled by crypto exchanges) usually have extremely high transaction volumes, tolerate excessive fees, and thus may set fees differently from other addresses. The final sample consists of 477,629 treatment and 594,395 control addresses[12] and a total of 21,827,333 transactions. Complete address-level data filtering process is discussed in Appendix D.

4.3 User (Address) Behavior Under Cap-Tip TFM

We now provide a descriptive understanding of how addresses make use of the two decision variables under the Cap-Tip TFM. It is important for two reasons. First, assuming senders and miners are rational and reasonably forward looking, Roughgarden (2020)'s game-theoretic analysis of EIP-1559 shows the equilibrium fee-setting strategy for senders includes a negligible Tip and a nontrivial gap between Cap and the actual fee paid. Second, the EIP-1559 transition adds information cues and decision flexibility for users. Understanding whether Cap or Tip is the primary decision variable at work helps us construct a hypothetical fee that reflects user-side welfare loss if only one decision variable can be used, as in the Legacy TFM. This hypothetical fee would, in turn, isolate the effect provided by the information cue.

As shown in Table 1, more than 75% of the transactions offer a Tip of 4.0 gwei or less, while more than half of the transactions have a Cap greater than 94.4 gwei. In addition, the gap between Cap and the actual fee is larger than 6.3 gwei for more than 75% of the transactions. These statistics show that the fee-setting behavior of addresses in reality matches the theoretical predictions to a reasonable extent. Furthermore, 15.34% of the transactions have Cap equal to the actual fee, which means, conversely, Tip is activated in 84.66% of the transactions. Appendix E plots the percentage of transactions with Tip or Cap activated by calendar day, demonstrating that Tip being the primary decision variable at work is a consistent pattern over time. The last row of Table 1 presents the gap between Cap and Base. The fact that Cap is much larger than Base implies that Cap is utilized as a safeguard mechanism by users.

4.4 Model Free Evidence for Overpayment Reduction After EIP-1559

To separate the effects of Base and Cap, we consider a hypothetical Tip-Only TFM where Cap is assumed to be muted, and hence the users pay Base + Tip instead of minCap, Base + Tip. Similar to the calculation of Cap-Tip overpayment in Section 4.2.1, we calculate a hypothetical Tip-Only overpayment, as illustrated in Appendix C. The hypothetical Tip-Only TFM is analogous to the Legacy TFM with the addition of Base, and hence the estimated Tip-Only TFM effect can be interpreted as the overpayment reduction attributed to Base.

Figure 3 presents the daily average overpayment among transactions using Legacy and Cap-Tip TFMs, as well as the

Table 1. User behavior under Cap-Tip transaction fee mechanism (N = 5,990,613).

Variable (gwei/gas)	Mean	Std.dev	First quartile	Median	Third quartile
Cap	123.83	255.44	64.72	94.43	132.30
Tip	19.24	128.46	1.50	1.57	4.00
Cap - Actual	39.31	206.42	6.34	18.92	37.34
Cap - Base	47.74	238.14	11.27	22.85	41.92

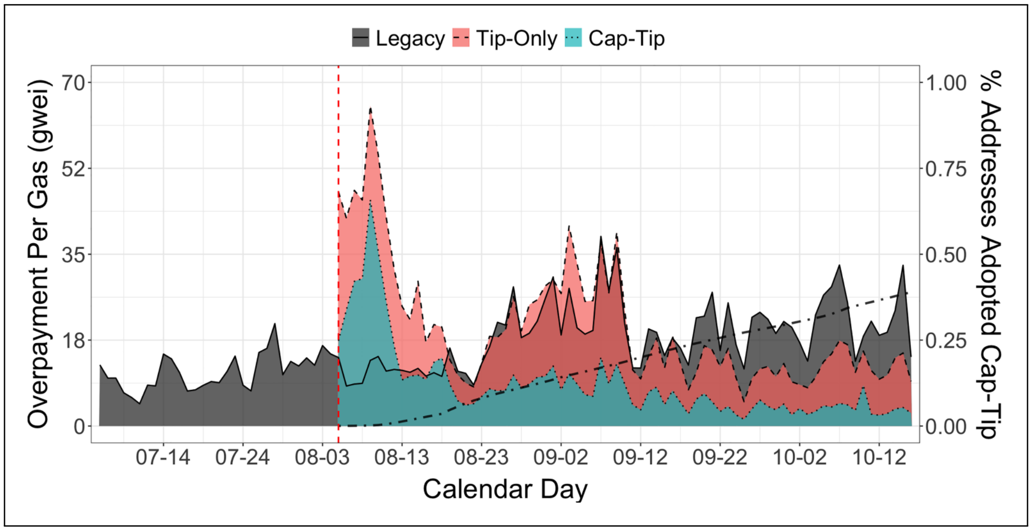

Figure 3. Daily average overpay amount and Cap-Tip TFM adoption rate.
Notes. This plot illustrates the daily changes in average overpayment under the Legacy (gray-shaded area with solid outline), the Cap-Tip (green-shaded area with dotted outline), and the hypothetical Tip-Only TFMs (red-shaded area with dashed outline), along with the proportion of active addresses using Cap-Tip TFM relative to the total number of active addresses (black dot-dashed line).

Cap-Tip adoption rate. Before the activation of EIP-1559 on August 5, 2021 (vertical dashed line), all transactions are Legacy (in solid curve). After August 5, 2021, the average overpayment among Cap-Tip transactions (in dotted curve) is consistently and often considerably lower than Legacy transactions, except for the initial two weeks. The average overpayment difference is 10.0 or 39.9%. Additionally, a fraction of the Legacy-to-Cap-Tip reduction is already captured by Tip-Only, especially in the later dates. Appendix E presents an alternative version of Figure 3 with daily total overpayment in US dollars. Combining both Legacy and Cap-Tip transactions, Ethereum users overpay an average of 4.6 million dollars per day, with a peak of over 11.6 million dollars (September 7, 2021).

4.5 *Identification Strategy*

Let $Y_{i,t}$ be the observed overpayment and $D_{i,t}$ the treatment status for address i on day t. If an address has multiple transactions on the same day, we take an average for overpayment and covariates and let $D_{i,t}$ be 1 as long as the last transaction uses Cap-Tip. We mark the initial policy date as $t = 0$ (i.e., August 5, 2021). It follows that the start date (July 6, 2021) and end date (October 16, 2021) of our sample period have $t = -30$ and $t = 72$, respectively (see Figure 4). All addresses have $D_{i,t} = 0$ for $t < 0$ and hence the untreated potential outcome, $Y_{i,t}(0)$, is observed. For $t \geq 0$, $D_{i,t}$ switches to 1 if address i is in the treatment group and has already adopted Cap-Tip on t. For clarity, the treatment addresses are further stratified by adoption time, g, where $g = 0$ if the first Cap-Tip transaction is observed on August 5, 2021 and $g = 42$ if that transaction is on September 16, 2021 (the last adoption date we consider). We correspondingly modify the typical treated potential outcome notation, $Y_{i,t}(1)$, into $Y_{i,t}(1, g)$. As a result, we have a total of 43 adoption-time groups with $g = 0, \ldots, 42$ in the treatment sample. For each group, the observation time window is ± 30 days around g (see an example in Figure 4). Thus, $Y_{i,t}(1, g)$ is observed on the $t \geq g$ days up to $t = g + 30$ and $Y_{i,t}(0)$ between $t = g - 30$ and $t = g - 1$ days.

Disaggregate ATTs. For each combination of adoption time (g) and post-adoption date ($t \geq g$), the corresponding average treatment effect on treated is defined by $ATT(g, t) = \mathbb{E}[\mathrm{Y_t}(1, \mathrm{g}) - \mathrm{Y_t}(0)]$. We implement a semiparametric approach introduced by Callaway and Sant'Anna (2021) to identify all $43 \times 31 = 1333$ ATTs.[13] Each ATT is estimated with a doubly robust estimator on a two-group by two-period subsample. This estimation method possesses the following three benefits: (1) it relies on the rather lenient conditional parallel trend assumption, (2) it avoids the negative weights issue suffered

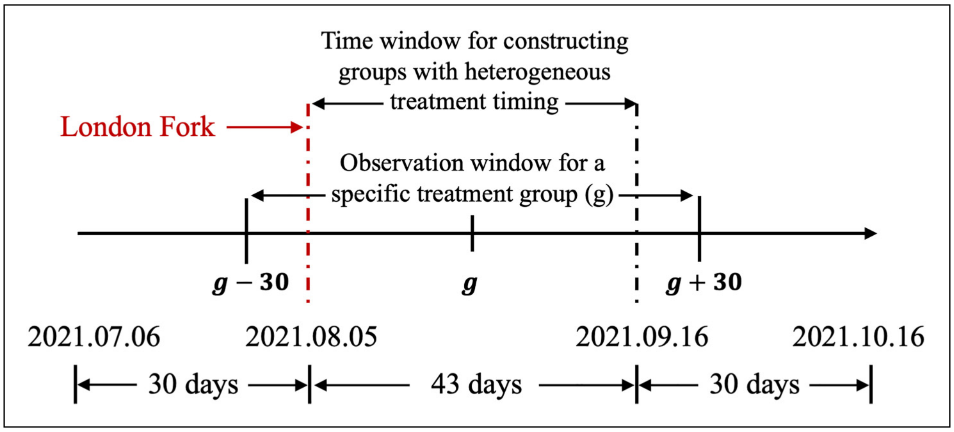

Figure 4. Treatment groups and observation time window by group.
Notes. This plot illustrates the construction process of the treatment group. The treatment group comprises addresses that adopt the Cap-Tip TFM between August 5, 2021 and September 16, 2021. For each treated address, a 30-day observation window is selected before and after the adoption.

by the two-way fixed-effect estimator, and (3) its flexibility of estimating various versions of the treatment effect integrates seamlessly with our research questions. See Appendix G for further discussion.

Aggregation. These granular ATTs are then aggregated to offer insights into the benefit of EIP-1559 in three ways. First, holding days after adoption ($e = t - g$) constant, aggregating $\widehat{ATT}(g, t)$ across the 43 groups weighted by respective subsample size produces a relative time estimate: $\widehat{ATT}_r(e)$, which informs about user learning as the exposure time e grows. Second, holding calendar time t constant, aggregating across groups that have already adopted Cap-Tip (i.e., $g \leq t$), again weighted by the subsample size, produces a calendar time estimate: $\widehat{ATT}_c(t)$, which informs how the benefit of EIP-1559 is related to daily system characteristics of the Ethereum network and the crypto sector at large. Last, aggregating across all g-t combinations forms an overall understanding of the short-to-medium-term average impact of the TFM update: $\widehat{ATT}_o$.

Parallel Trend Assumption. Despite being granular and doubly robust, our estimator relies on the same parallel trend-identifying assumption as in other DiD estimators, which is not testable directly (Callaway and Sant'Anna, 2021: 204–205). Following recommendations in recent econometric reviews (Kahn-Lang and Lang, 2020), we perform a commonly used pre-treatment parallel trend test using our estimator in Section 5.2. As a side note, the validity of the stable unit treatment value assumption is discussed in Appendix J.

Conditional Exogeneity of Treatment Timing. A typical Ethereum user manages their crypto assets and initiates transactions through "wallets." After the EIP-1559 rollout on August 5, 2021, wallets had to release a new software version supporting the Cap-Tip TFM.[14] Users of a wallet are then able to upgrade their software version to use the new TFM. The process of updating the wallet is much like updating a browser plug-in, which takes a minimal amount of time. Thus, the timing of receiving the treatment is related to the user's reliance on wallets. This reliance might also affect overpayment via the wallet's fee recommendations and, hence, if not controlled for, might give rise to omitted-variable bias. To this end, we construct a proxy variable for wallet reliance, explained further in Section 4.6. Given the level of wallet reliance, users are unlikely to strategize the timing of wallet updates based on their preference between the Legacy and the Cap-Tip TFMs.[15] Rather, when to update is most likely driven by one's time availability before initiating the next transaction. In our view, such contemporaneous availability is highly random and perhaps at best weakly correlates with the activeness of an address. This leads us to also construct a set of activeness-related covariates. One might also argue that market conditions dictate one's urgency to trade and hence the time availability before the transaction. However, all time-related factors are inherently controlled for in our estimator as for each $\widehat{ATT}(g, t)$, the treated and control addresses experience the same market conditions. To summarize, we consider treatment timing exogenous conditional on our set of control variables.

4.6 Variable Definition and Summary Statistics

We construct four sets of covariates. First, to capture the activeness of address i prior to day t, we open a 6-month window from t-180 to t-1 and collect all transactions initiated therein. Then, we calculate the number of active days (ActDays6M), the average transaction value (TxValue6M), the average address balance (Balance6M), and the number of transaction types (TxType6M)[16] using these transactions. Second, treated and control addresses might initiate transactions at different times within a day. In a large enough g-t

subsample, this difference is unlikely to be systematic. Nevertheless, we compute two fee-per-gas measures with intraday granularity and control them for the potential benefit of estimating certain ATTs in small subsamples. Specifically, AvgMedGasPrice15B averages the within-block medium fee per gas across the 15 blocks prior to the focal transaction. AvgMedGasPrice8H divides a day into three 8-hour segments and calculates medium fees per gas across all blocks in the same segment as the focal transaction. Third, we also measure the focal transaction's value (TxValue) and the total amount of gas used (TxGasUsed) as higher-value transactions might have a higher overpayment tolerance.[17] For the second and third sets of covariates, if an address has multiple transactions on day t, they are averaged to the address-day level. Fourth, users might exhibit varying degrees of reliance on the wallet-recommended fees, which could, in turn, correlate with their fee-setting performance. Although we do not directly observe such user reliance, we create a proxy variable, RecReliance6M (see exact procedure in Appendix H), by exploiting the fact that wallet-recommended fees stay constant between two blocks. Hence, if a focal user frequently sets fees at the same level as transactions in their own and adjacent blocks, their wallet reliance must be high.

Transactions in our data are aggregated into 6,974,141 address-day level observations; see Appendix I for variable definitions and statistics. Log transformation is applied to overpayment, overpayment (Tip-Only), TxValue, TxGasUsed, ActDays6M, Balance6M, and TxValue6M.

5 Empirical Analyses and Results

We organize our results as follows: Section 5.1 discusses the overall and decomposed effects of EIP-1559 (Hypotheses 1 and 2); Section 5.2 examines heterogeneity in exposure time; Section 5.3 explores heterogeneity across small and large addresses (Hypotheses 3a and 3b); Section 5.4 addresses calendar day heterogeneity estimated with post-learning effects; and Appendix K reports robustness checks.

5.1 Overall Impact of EIP-1559 on Reducing User-Side Welfare Loss

We carry out the estimation procedure described in Section 4.2 on both Cap-Tip overpayment and hypothetical Tip-Only overpayment, each producing a set of disaggregate average treatment effects: $\widehat{ATT^k}(g,t)$, where $k = \{CT,\ TO\}$ indicates the corresponding dependent variable. We aggregate $\widehat{ATT^k}(g,t)$ across both adoption group and exposure time ($e = t - g$) dimensions to form the overall impact of the EIP-1559 transition: $\widehat{ATT_O^k} = \sum_{g=0}^{42}\sum_{e=0}^{30} w_{g,t}\widehat{ATT^k}(g,t)$, where the weight, $w_{g,t} = N_{g,t}/N$, is the sample size proportion of the g-t subsample over the full sample (Callaway and Sant'Anna, 2021: 221). As a benchmark, we also present results from the common two-way fixed-effect estimator. Due to its well-documented issue of negative weights in the staggered adoption settings (de Chaisemartin and D'Haultfoeuille, 2020), we do not use this method in the remainder of the analysis. Results are presented in Table 2.

Our estimates indicate that the log(Overpayment) reduces by −1.137 on treated addresses as they switch from the Legacy to the Cap-Tip TFM, i.e., $\widehat{ATT_O^{CT}} = -1.137$. To interpret this effect on a percentage scale, we use $\widehat{ATT_O^{CT}}$ to predict a counterfactual overpayment if a treated address had used the Legacy TFM instead. Then, the percentage reduction in Cap-Tip overpayment over the counterfactual Legacy overpayment is averaged across the sample. As shown in Table 2 (the third row), the welfare loss improvement of Cap-Tip is substantial: a reduction of 81.45% in user overpayment of transaction costs. Using a \$3000/ETH exchange rate this translates into \$3.29 per transaction and over \$3.96 million in system-wide daily savings at the full adoption rate. As predicted by H1 and H2, the two changes enacted by EIP-1559, Base, and Cap, should benefit user-side welfare separately. The Tip-Only results in Table 2 provide a direct test of H1. If we hypothetically disable Cap, analogous to setting Per-gas Fee (Legacy) in the presence of Base, the log(Overpayment) reduction, $\widehat{ATT_O^{TO}}$, is estimated at −0.796 ($p < .001$), which translates into the hypothetical Tip-Only mechanism having a 71.07% lower overpayment than the Legacy counterfactual. Thus, H1 is supported: Base functions as a valuable congestion signal for Ethereum users when setting their transaction fees. The difference between the Cap-Tip and the Tip-Only ATTs, $\widehat{ATT_O^{CT}} - \widehat{ATT_O^{TO}} = -0.341$ ($p < .001$), is the impact of adding Cap to the decision-making process. This supports H2: the dual-variable Cap-Tip TFM forces users to decompose an otherwise challenging decision task, reducing user-side welfare loss.

The above results further allow us to quantify the relative contributions of Base and Cap. This is shown in the last column of Table 2: 70.07% and 29.93%, respectively. While the congestion signal, Base, plays a larger role in reducing user-side welfare loss, it is interesting to note that Cap is activated in only 10–20% of transactions. This implies that when Cap works, it avoids serious welfare loss. To verify, we calculated the difference between Cap and Base + Tip for each Cap-activated transaction, which measures how much the address's overpayment would have increased without Cap. The average of this difference is 70.5 gwei. In contrast, the sample mean of overpayment is only 8.3 gwei, providing descriptive evidence for the conjecture that "when it works, it saves big."

5.2 User Learning of the New TFM

Knowing the overall effect EIP-1559 imposes on user welfare, a natural follow-up question is how fast users learn to attain this benefit. Furthermore, as will be evident in Section 5.3, exploring how the decomposed effects of Base and Cap evolve post adoption can be particularly helpful for comparing and interpreting the differences between small and large users.

Table 2. Overall impact of EIP-1559 on Ethereum user transaction fee overpayment.

	Cap-Tip		Tip-Only		Contribution of Base vs. Cap
	$\widehat{ATT}_O^{CT}$	Cf %	$\widehat{ATT}_O^{TO}$	Cf %	$[\frac{\widehat{ATT}_O^{TO}}{\widehat{ATT}_O^{CT}}, 1 - \frac{\widehat{ATT}_O^{TO}}{\widehat{ATT}_O^{CT}}]$
CSDiD	−1.137***(0.044)	−81.45%	−0.796***(0.046)	−71.07%	[70.07%, 29.93%]
TWFE	−0.995***(0.009)	−77.90%	−0.643***(0.013)	−64.81%	[64.67%, 35.33%]

Cf% = counterfactual %. Robust standard errors in parenthesis. N = 6,974,141. The second row presents the results estimated using a two-way fixed effects (TWFE) model applied to the matched dataset (N = 3,759,496). The matching process was conducted through nearest-neighbor dynamic matching with a caliper of 0.05, based on the propensity score. The matching variables included: Log(1 + Balance6M), Log(1 + TxValue6M), Log(1 + ActDays6M), TxType6M, and RecReliance6M.
$^{***}p < .001$.

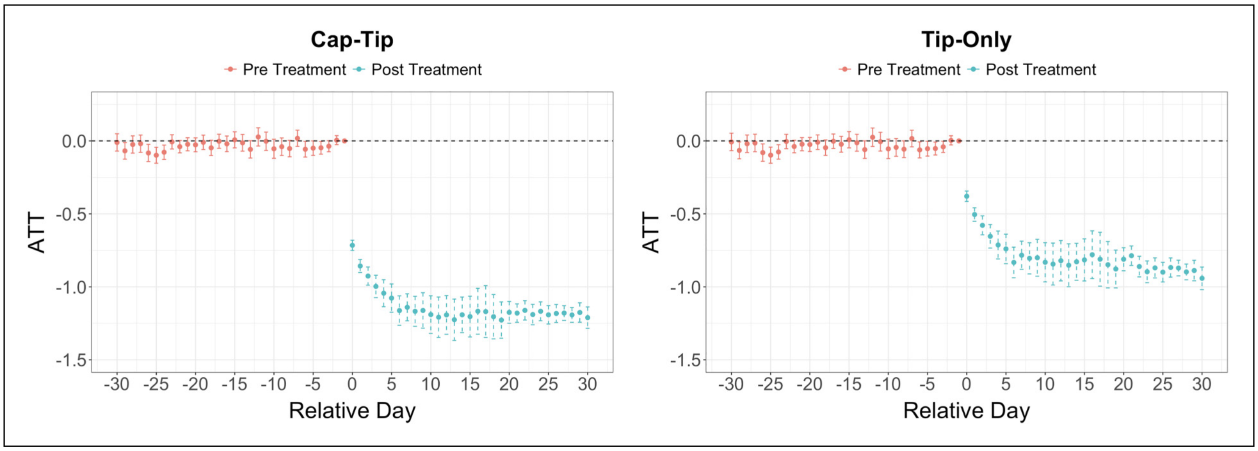

Figure 5. Impact of EIP-1559 on overpayment by exposure time.
Notes. These plots illustrate how the average treatment effect on the treated (ATT) varies with the exposure duration, using Cap-Tip overpayment (left panel) and Tip-Only overpayment (right panel) as the dependent variable. Since each treated address is observed for a time window of 30 days before and after EIP-1559 adoption, we estimate a total of 30 pre-treatment ATTs (in red solid error bars; ATT = 0 for e = −1 period by design) and 31 post-treatment ATTs (in green-dashed error bars; e from 0 to 30). Pre-treatment ATTs are estimated using t and "g − 1" periods.

As a result, we defer testing Hypotheses 3a/3b to the next section. Firstly, H1 posits that Base provides more complete and current congestion information to users compared with the pre-EIP-1559 era. Secondly, H2 argues that adding Cap forces task decomposition in a meaningful way. Thus, ease of use underlies both hypotheses, implying that the learning speed should not be slow. We aggregate $\widehat{ATT}^k(g,t)$ to the relative day level: $\widehat{ATT}_r^k(e) = \sum_{g=0}^{42} w_{g,e}\widehat{ATT}^k(g, g+e)$, where $k = \{CT, TO\}$ and $e = t - g$ are the number of days exposed to the treatment (i.e., relative days). The weight $w_{g,e}$ is the sample size proportion as defined above. Following Callaway and Sant'Anna (2021: 218)'s suggestion, we use pre-adoption estimates (i.e., $\widehat{ATT}_r^k(e)$ when $e < 0$) to test for parallel trend and assume the same trend persists after adoption. Results are shown in Figure 5. For both Cap-Tip and Tip-Only, the pre-adoption ATTs are closely oscillating around zero (red solid error bars in figure), offering strong evidence for parallel trends.

We observe significant user learning at least in the first 6 days after adoption (green dots with dashed standard error bars in Figure 5). For example, the Cap-Tip ATT starts at -0.72 on $e = 0$ and decreases to -1.16 on $e = 6$. Six days post adoption correspond to 12 transactions for a typically treated address. Similar conclusions regarding learning can be drawn on $\widehat{ATT}_r^{TO}(e)$. In online experiments with repetitive tasks, Goldstein et al. (2020: 1380–1381) find that it takes no more than seven rounds for subjects to reach the optimal solution in a simple search task, while Van den Bos et al. (2008: 486) report a much slower learning process (over 50 rounds of bidding) in a small-scale auction in lab conditions. Thus, given the task complexity of setting transaction fees in a nuanced real-world environment, we consider the user's adaptation to the Cap-Tip mechanism rapid. In addition, the entire Tip-Only curve for $e \geq 0$ resembles an upward shift of the Cap-Tip. This implies user learning happens primarily on setting Tip, while the impact of Cap on overpayment reduction stays somewhat stable across varying exposure times. This result aligns well

Table 3. Impact of EIP-1559 on transaction fee overpayment by the user group.

		Cap-Tip		Tip-Only		Contribution of Base vs. Cap
	N	$\widehat{ATT}_O^{CT}$	Cf%	$\widehat{ATT}_O^{TO}$	Cf %	$[\frac{\widehat{ATT}_O^{TO}}{\widehat{ATT}_O^{CT}}, 1-\frac{\widehat{ATT}_O^{TO}}{\widehat{ATT}_O^{CT}}]$
Small Adr.	5,920,297 (84.89%)	−1.162***	−82.21%	−0.849***	−73.18%	[73.02%, 26.98%]
Large Adr.	1,053,245 (15.11%)	−0.962***	−75.26%	−0.427***	−50.25%	[44.41%, 55.59%]

Robust standard errors in parenthesis. Adr = address; Cf% = counterfactual %. Disaggregate ATTs are not estimated on g-t subsamples with N ≤ 10 as sample size needs to be larger than number of covariates in X. Seventy-four subsamples (599 observations) for large addresses are dropped as a result.
***$p < .001$.

with our theoretical argument that Cap and Tip, as two separate decision variables, force task decomposition. Cap-setting is largely driven by internal factors such as willingness-to-pay, which is more straightforward and, as it turns out, does not require much learning.

5.3 Heterogeneous Impact of EIP-1559 on Small Versus Large Addresses

As discussed in the Introduction, despite permissionless blockchain's decentralization of access, a clear benefit for the underbanked population, the Legacy TFM favors resource-rich large addresses in fee estimation. Encouragingly, descriptive evidence in Figure 1 and theorization in H3 both point to higher user-side welfare benefits for small addresses brought by EIP-1559. We now tackle this question empirically. During the analysis window, addresses are categorized by the average transaction value. Those above (on or below) \$5000 are labeled as large (small) addresses, yielding a roughly 15%:85% sample size ratio.[18]

Results are shown in Table 3. The overall reduction in overpayment, $\widehat{ATT}_O^{CT}$, is estimated at −1.162 for small and at −0.962 for large addresses. The percentage version of $\widehat{ATT}_O^{CT}$ (under Cf % in Table 3) is −82.21% for small and −75.26% for large addresses. Overall, EIP-1559 benefits small addresses more than large addresses. This difference is statistically significant,[19] supporting H3a. Next, similar to Section 5.1, we use the Tip-Only result, $\widehat{ATT}_O^{TO}$, to attribute EIP-1559's overall impact to its two elements: Base and Cap. Interestingly, small addresses are noticeably more effective at utilizing the Base information to set an appropriate Tip, resulting in lower overpayments. $\widehat{ATT}_O^{TO}$ is equal to −0.849 and −0.427 for small and large addresses, respectively. This is in line with H3b, which states that large addresses are more likely to suffer from overconfidence bias and carry their Legacy fee-setting habit over to Tip, resulting in an unnecessarily large Tip and higher overpayments.

Next, we examine whether the two user groups exhibit heterogeneity in adapting to the Cap-Tip TFM, following the approach outlined in Section 5.2. Results are shown in Figure 6. The post-adoption learning process ($e \geq 0$) for small addresses is almost identical as in the full sample, i.e., the top rows of Figure 6 and Figure 5 are highly comparable. However, interestingly, large addresses experience a much slower learning process. For the first 20 days post exposure, $\widehat{ATT}_r^{CT}(e)$ gradually decreases from −0.67 to roughly −1.06. It somewhat stabilizes for $e \geq 20$, albeit being bumpier than the small address curve, which is not unexpected given the smaller sample size. $\widehat{ATT}_r^{TO}(e)$ exhibits a similar learning pattern, which again is in line with our theorization on the behavioral bias of large users in H3b.

Last, a comparison between Figure 6(a) and (c) reveals that although $\widehat{ATT}_r^{CT}(e)$ for both small and large addresses converges to a similar level at the end of the exposure window, the slower learning process of large addresses makes their overpayment-reduction effect smaller in the initial days post adoption. In summary, we find strong evidence that large addresses face poorer Tip-setting performance, consistent with the theoretical prediction of experience-induced overconfidence. This luckily propagates only to a limited degree to the actual overpayment (i.e., Cap-Tip performance), as coincidentally, large addresses also tend to set a Cap that is closer to Tip in the initial stage post exposure. This behavior is consistent with large addresses initially having a less clear task decomposition.

5.4 Does System Congestion Tamper With the Effectiveness of EIP-1559?

We now turn to the question of whether Cap-Tip is less effective under certain conditions. Since the role of Tip is to compete with other users for the limited block space, it is conceivable that users are tempted to set an unnecessarily high Tip when they perceive intense system congestion. This conjecture leads Ferreira et al. (2021) to question the advantage of Cap-Tip over Legacy. The availability of Base as a contemporaneous congestion signal might further exacerbate this concern.

To ensure a clean investigation of the above perspectives, we remove the learning effect by focusing on *g-t* groups with $e \geq 20$. This means our observation window on a calendar day scale is between August 25 and October 16, 2021, where the former is exactly 20 days after the earliest adoption. This time window has the additional benefit of removing the

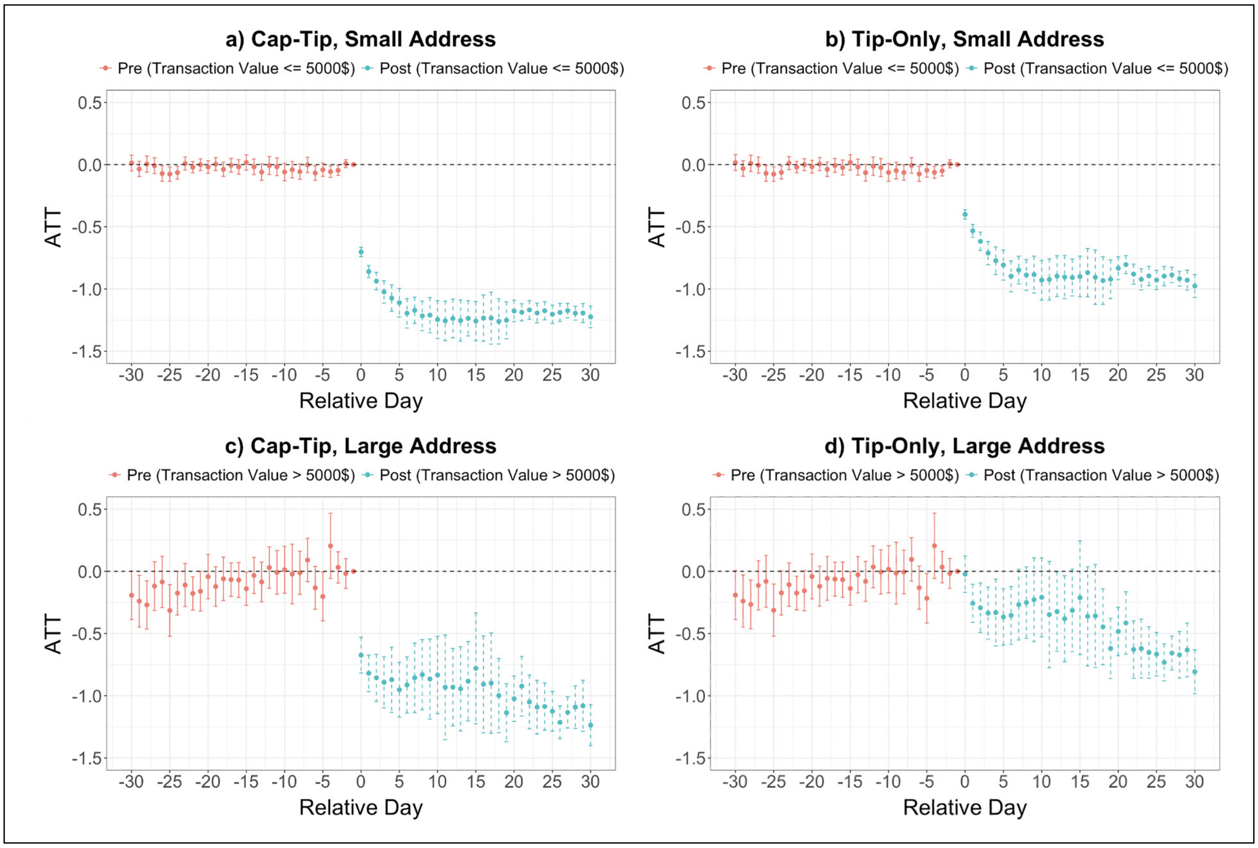

Figure 6. ATT by exposure time: small (top) versus large (bottom) addresses.
Notes. These plots repeat Figure 5 in small and large address subsamples. Addresses with average transaction value below/above \$5000 are labeled as small/large addresses, yielding a roughly 85%:15% small-large sample size ratio.

potential impact of a MetaMask (most popular ETH wallet) update related to fee-setting on August 18. Then, we aggregate relevant ATTs to the calendar day level: $\widehat{ATT}_c^{CT}(t) = \sum_{g=\max\{t-30,0\}}^{\min\{t-20,42\}} w_{g,t}\widehat{ATT^{CT}}(g,t)$. Figure 7 (left panel) shows the aggregation results and daily average Base. Surprisingly, contrary to the suspicion in the literature, the welfare-improving performance of Cap-Tip is stronger during higher system congestion, evidenced by the clear negative correlation (-0.14) between daily Base and $\widehat{ATT}_c^{CT}(t)$. In other words, the concern that Tip carries the drawbacks of a first-price auction and leads users to suffer as they do with the Legacy Per-gas Fee does not materialize. To verify, we plot daily Cap and Tip (within the $e \geq 20$ sample). While Cap moves in almost perfect tandem with Base (correlation $= 0.34$), Tip stays at a very low level, as shown in Figure 7 (right panel). In fact, the daily medium Tip has never exceeded three gwei and is negligible on most days. Recall from Section 4.3 regarding the user's theory-consistent fee-setting behavior. The above evidence extends this conclusion from the sample average level to the calendar day level. Thus, after initial learning, users are able to exercise Cap-Tip fee-setting in a near-optimal way. We suggest this somewhat surprising yet encouraging result is partly due to the task decomposition benefit of separately setting Cap and Tip, as posited in H2.

6 Conclusion

Decentralized permissionless blockchains are often touted with strong security and radical transparency. Yet, such benefits of decentralization do not come free. From the user's perspective, the transaction fee is the most salient cost for maintaining a blockchain system. This study leverages the EIP-1559 update on the Ethereum to examine the user-side welfare impact of the two most widely adopted TFMs, answering Cachon et al. (2020)'s call for research on how system mechanism design affects user welfare and Mithas et al. (2022a)'s broader call for empirical blockchain research.

Our findings show that the Cap-Tip TFM substantially reduces user-side welfare losses by approximately 80%, attributable to the combined influence of the Base and Cap components. Notably, Base accounts for nearly 70% of the impact, reflecting its role in enhancing information quality, specifically in terms of completeness and currency. This highlights despite data transparency being blockchain's inherent feature, how information is compiled and presented to users

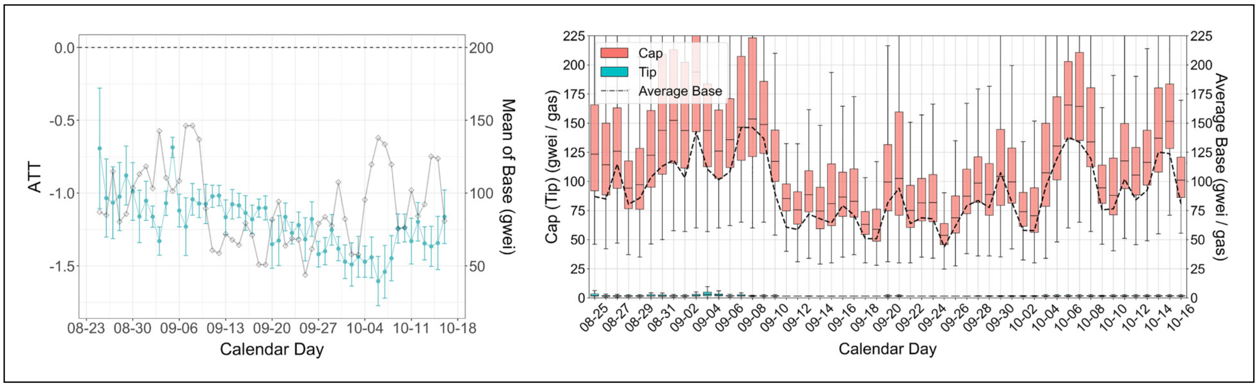

Figure 7. Calendar day ATT (left), Cap and Tip (right).
Notes. The left panel illustrates how the average treatment effect on the treated (ATT) varies with calendar days (green dots with error bars) alongside the daily mean value of Base fee (gray solid line). The right panel presents a boxplot depicting the daily distribution of the Cap (red boxes) and Tip (green boxes) values and the daily mean value of the Base fee (black dashed line) for transactions in our sample.

remains critical. Meanwhile, Cap contributes to the remaining 30% of the effect by enabling an effective task decomposition for users. Furthermore, we find that, on average, small users require approximately 6 days to fully realize the welfare benefits, with the learning process focused mostly on setting Tip. Large users, surprisingly, take longer to adapt due to overconfidence, echoing prior findings on experienced users/experts discounting algorithms or system-provided recommendations (Logg et al., 2019; Soll et al., 2022). Interestingly, the different reaction of small and large users results in greater fairness within the transaction fee market, adding to the broader literature on fairness in operational designs (Bertsimas et al., 2012). Finally, contrary to theoretical predictions (Reijsbergen et al., 2021), we show that the Cap-Tip TFM performs even better during periods of high congestion.

Out study offers three key practical implications for blockchain operators. First, although transparency is often highlighted as a key benefit of blockchain systems, accessing on-chain information and utilizing it for decision-making (e.g., setting transaction fees) remains challenging. This challenge inherently grants the resource-rich large users an advantage over smaller ones, potentially resulting in inequity. Therefore, providing users with additional easy-to-use and easy-to-access information within blockchain-based ecosystems remains crucial. The Base component in the Cap-Tip TFM serves as an effective example. In other blockchain contexts, such as in the proposal voting process of DAOs (see Zhao et al. (2022) for a primer), information provision policies can be designed with similar motivations in mind. For instance, when DAO voting offers a delegation option, displaying information related to the delegate has the potential to keep new and small users better-informed in making their choices.

Second, despite that ease-of-use considerations are embedded in the design principles of the Cap-Tip TFM, the learning period is still nontrivial, as we show in Section 5.2. Hence, blockchain policymakers should prioritize educating users on how to effectively utilize new mechanisms before their implementation. This can be achieved through increased outreach on mainstream social media platforms and official websites.

Third, given the sizeable reduction of transaction fee overpayment we demonstrate, it is natural to ask whether the success of Cap-Tip TFM on the Ethereum blockchain is generalizable to other permissionless blockchains. We believe the following characteristics of Ethereum are the prerequisite for Cap-Tip's success and hence are important criteria to consider for generalizability: (1) relatively short inter-block time, (2) diverse user base with varying sensitivities to transaction fees, (3) sufficient protocol incentives for miners/validators compared to their operational costs, and (4) contemporaneous demand frequently exceeds supply. We conduct a qualitative review of the Top 15 permissionless blockchains ranked by volume and find that one-third (e.g., Solana, BSC, and Polygon) share these four characteristics. In the following, we briefly outline why a blockchain that deviates from any of these four characteristics may fail to realize the benefits of adopting the Cap-Tip TFM.

To begin with, inter-block time (i.e., time elapsed between two adjacent blocks) determines how quickly the Base fee can be updated. Thus, for blockchains with long inter-block times such as Bitcoin, the Base fee becomes less capable of approximating real-time system congestion. This in turn forces users to rely more on Tip to differentiate from others and deviates from the design principle of the Cap-Tip TFM. Next, since the dynamic Base fee acts as a filter for users with varying sensitivities to transaction fees, the Cap-Tip TFM is more suitable in blockchains with diverse user demographics. For example, traders primarily engaging in decentralized finance (DeFi) transactions are likely more sensitive to transaction speed than cost, compared to general users. Thus, for blockchains (e.g., Mixin, Gnosis, and Tron) with narrowly focused transaction types, such as microtransactions, decentralized governance, niche NFTs, and prediction markets, the effectiveness of the Cap-Tip TFM may be limited. Furthermore, as the Base fee is burned per the Cap-Tip TFM, the remaining transaction fee rewards, i.e., minTip, Cap-Base, must provide enough incentive for miners/validators to participate. From this perspective, the Cap-Tip TFM is more suitable for PoS than PoW blockchains, since computational costs for mining are a lot higher in the latter. Last, both Legacy and Cap-Tip TFMs are solutions to the blockchain-based congested service system where user sets the priority fee. These solutions are relevant only if the system is frequently congested. We find that, as of March 2025, several mainstream blockchains, including Layer 1 solutions such as Fusion, Cronos, Avalanche, Kava, and Tron, and Layer 2 solutions such as Arbitrum,[20] Optimism, Mixin, and Gnosis, do not experience congestions. In such scenarios, simpler mechanisms such as first-come-first-served (FCFS) or fixed/posted price may suffice and could even enhance user experience.

In summary, the above discussion suggests that a one-size-fits-all policy, such as adopting Cap-Tip TFM for all permissionless blockchains, is likely too simplistic for policymakers contemplating transaction fee regulations. For blockchains that do not meet any of the four criteria, the Cap-Tip TFM may underperform relative to its observed success on Ethereum and could even potentially disrupt the incentive structure at the consensus layer (e.g., in PoW systems). Therefore, operators of blockchain platforms should thoroughly assess the suitability before implementing the Cap-Tip TFM. For emerging blockchain platforms, in particular, it is crucial to design TFMs in alignment with the platform's consensus mechanism, scalability objectives, business ecosystem structure, and long-term strategic roadmap.

To close the study, we discuss several limitations that could lead to future research. First, during the EIP-1559 update, Ethereum's consensus mechanism was still PoW. The consensus change to PoS happened over 1 year later, which is too far out for our analysis. We have no a priori reason to believe the consensus change will alter our conclusions on Cap-Tip's performance over Legacy since the difference between PoS and PoW lies in the rules determining who has rights to assemble and verify a new block and the corresponding miner/validator incentives, which are rather independent from user's transaction fee-setting behavior and hence overpayment. Nevertheless, future research could consider investigating an update similar to EIP-1559 in a different PoS blockchain. Second, due to the anonymous nature of permissionless blockchains, we can only measure user characteristics with their on-chain activity, making it challenging to contextualize users as off-chain real-life characters. With the increasing adoption of de-anonymization applications such as Ethereum Name Services, future research can map addresses with social media data and differentiate users beyond on-chain activities such as small versus large users.

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Notes

1. As another branch, permissioned blockchains (e.g., Hyperledger Fabric) are the closed version of permissionless blockchains, where participation in some or all roles requires user approval. Our discussion in this study focuses on the permissionless branch. Due to its decentralized nature, a consensus mechanism is required. The two main mechanisms are PoW, used by Bitcoin, and PoS, used by Ethereum post-15 September 2022. Our data was collected during Ethereum's pre-PoS period, and we will thus refer to network maintainers as miners in the following.
2. See Ilk et al. (2021) and Shang et al. (2023) for a detailed description of the mining process on Bitcoin. Ethereum, prior to the PoS transition, shared a highly similar process.
3. See https://coinmarketcap.com/academy/glossary/blockchain-trilemma for more details.
4. Based on data accessed on 1 March 2023 from https://statoshi.info/ and https://etherscan.io/.
5. See, for example, https://www.ig.com/it-ch/prime/insights/news/what-are-the-advantages-of-algo-trading-220127.
6. Note that it is difficult to accurately estimate the value included in contract transaction (e.g., NFT transfers), we only consider the normal ETH-transfer transactions in the 3 months before and after the London Hard Fork and calculate the USD value. Specifically, we select 4 May 2021 to 4 August 2021 and 6 August 2021 to 6 November 2021 as our observation windows.
7. Ethereum is a mainstream permissionless blockchain network that allows users to build decentralized apps, create smart contracts, transfer fungible and nonfungible tokens (NFTs), and vote in a DAO. Regardless of the purpose of an Ethereum transaction, it is imperative for the user initiating the transaction (i.e., the sender) to offer a transaction fee to miners. The transaction is generally considered "final" or immutable in the blockchain glossary, some time after it goes on-chain. How long this time

should be is an open debate. In Bitcoin, for example, most people believe once a transaction is included in a block and at least six additional blocks are chained to the block, the transaction is called “confirmed.” This takes about 60 minutes. See https://www.coincenter.org/education/crypto-regulation-faq/how-long-does-it-take-for-a-bitcoin-transaction-to-be-confirmed/ for more details. Ethereum takes considerably shorter time to confirm a transaction. The ideas inside EIP-1559 were first made public in Buterin et al. (2019). It was extensively discussed in the Ethereum developer community and considered one of the most significant updates to the Ethereum network (see https://notes.ethereum.org/@vbuterin/eip-1559-faq for more details). The implementation of EIP-1559 was wrapped inside the London Hard Fork on 5 August 2021, together with a few other relatively minor updates. Refer to https://ethereum.org/en/history/#london for more details on the London Hard Fork.

8. Fee is measured by gwei. A gwei is one-billionth of one ETH. Gas is a concept native to Ethereum, which refers to the basic unit of computational resources on Ethereum. Size and capacity on Ethereum are then measured in units of gas. One can draw analogy of gas to the more familiar concept of data space. An Ethereum transaction consuming a certain amount of gas is similar to a digital file occupying a certain amount of data space, such as in bytes. In practice, when setting Per-gas Fee, the sender also decides a limit on the gas they are willing to consume. The transaction fails if the actual needed gas is more than this limit. If instead the limit is higher, the sender only pays for the gas consumed. Hence, almost all transactions will set a high enough limit.
9. Note that, the “Base” portion of transaction fees is burned instead of given to miners. However, since minting rewards typically constitute the majority of miners’ revenue (Capponi et al., 2023; Pagnotta, 2022)—minting-to-fee revenue ratio is 4:1 in our data—such burning does not fundamentally change miners’ incentive to participate. Interested readers are referred to the documentation of EIP-1559 (Buterin et al., 2019) for additional details on its potential implications for miners and the circulation of the ETH cryptocurrency.
10. To our knowledge, during our research period, the most widely used Ethereum transaction information aggregator, Etherscan, and the most popular wallet, MetaMask, did not undergo any significant layout or information display changes.
11. Broadly speaking, user (or consumer) welfare indicates the benefits a user obtains from the consumption of goods or services (Brynjolfsson, 1996). Two approaches exist in the literature: experiments and model-based counterfactuals. The former asks subjects to engage in a hypothetical purchase situation and directly elicits their product valuation (e.g., Hinz et al., 2011). The latter assumes a theoretically grounded consumer utility model, fits it to data, and uses the estimated model to compute consumer welfare via counterfactual analysis (e.g., Feldman et al., 2022). Thus, the former approach enjoys the benefit of a direct measure (instead of a model-based counterfactual measure), while the latter has the advantage of relying on actual consumer decisions (instead of using vignettes). Our measure is based on actual user decisions yet circumvents the need for specifying an underlying utility model.
12. To be rigorous, we note that one cannot identify users on Ethereum but only addresses control by users (Wang et al., 2021). In other words, if a user controls several addresses, they cannot be linked by a common user ID. As a result, our main empirical analysis is performed at the address level. As long as users exhibit consistent behavioral patterns across the addresses they control, our estimates can also be interpreted at the user level.
13. In execution, if a specific *g-t* subsample has data sparsity in either the treatment or the control group, the estimator might not converge. We report the number of disaggregate ATTs dropped under each results table in Section 5.
14. For example, MetaMask, a mainstream wallet, released the Cap-Tip compatible version on 10 August 2021 (see Figure F-1 and Figure F-2 in Appendix F).
15. One can at least enjoy the information cue benefit of the latter by setting Cap equal to Base + Tip and effectively nullifying the one extra decision variable.
16. The two main types on the Ethereum blockchain are token transfer transactions and smart contract transactions. The latter can be further categorized by the type of Dapp the contract is referencing, including Finance, NFT-related, Gaming, Gambling, DAO-Voting, Social, and Others (Dapp categorization by DappRadar).
17. TxValue and TxGasUsed are distinct concepts: while the former captures the amount of token transferred from one address to another, the latter is determined by transaction complexity. Correlation between the two variables is indeed very low, 0.006.
18. When dichotomizing Ethereum users into large versus small groups, we face a trade-off between the precision of identifying large users and the data sparsity for large users. The latter in turn influences how well our DiD estimates for the large-user group can satisfy the parallel trend assumption. The *AvgTxValue* = $5,000 cutoff corresponds to top 10.7% users who together contribute 95% of total transaction value on Ethereum. Thus, they are likely representative of the large users who would truly behave differently from the rest of the user population, if Hypothesis 3a and 3b hold. The resulting transaction sample size (15%) also ensures our DiD estimation can reasonably satisfy the the pre-treatment parallel trend test.
19. Testing the significance of ATT difference across small and large addresses under the split sample analysis relies on an implausible “sample independence” assumption. Thus, we resort to an imputation-based approach by Borusyak et al. (2024), which estimates a triple difference model and tests an overall large-versus-small ATT difference, which is significant at the 0.001 level. In general, heterogenous treatment effect based on unit-traits under the staggered adoption setting is still an active area of econometric research. Empirical research in OM/IS (e.g., Mayya and Huang, 2025) typically apply the two-way fixed-effect model, despite its known negative weights issue. Recent applied economics research (Cao et al., 2024), similar to us, also make use of Borusyak et al. (2024).
20. Interestingly, while Arbitrum adopts the Cap-Tip TFM, its official documentation indicates that a typical transaction only pays the Base fee, with the Tip inactive, i.e., set to zero. This highlights that the Cap-Tip TFM is unnecessary in systems with excess supply. We believe the adoption of Cap-Tip by many Layer 2 blockchains is not due to necessity but due to convenience (for developers). Unfortunately, this imposes unnecessary decision-making burden on users.

References

Andreasson K (2022) Digimentality 2022—Fear and favouring of digital currency. Retrieved March 22, 2023. Available at: https://impact.economist.com/projects/digimentality-2022/.

Babich V and Hilary G (2020) OM Forum—distributed ledgers and operations: What operations management researchers should know about blockchain technology. *Manufacturing & Service Operations Management* 22(2): 223–240.

Bapna R, Jank W and Shmueli G (2008) Consumer surplus in online auctions. *Information Systems Research* 19(4): 400–416.

Basu S, Easley D, O'Hara M, et al. (2023) Stablefees: A predictable fee market for cryptocurrencies. *Management Science* 69(11): 6508–6524.

Bertsimas D, Farias VF and Trichakis N (2012) On the efficiency-fairness trade-off. *Management Science* 58(12): 2234–2250.

Billett MT and Qian Y (2008) Are overconfident CEOs born or made? Evidence of self-attribution bias from frequent acquirers. *Management Science* 54(6): 1037–1051.

Bolandifar E, DeHoratius N, Olsen T, et al. (2019) An empirical study of the behavior of patients who leave the emergency department without being seen. *Journal of Operations Management* 65(5): 430–446.

Borusyak K, Jaravel X and Spiess J (2024) Revisiting event-study designs: Robust and efficient estimation. *The Review of Economic Studies* 91(6): 3253–3285.

Browne GJ, Pitts MG and Wetherbe JC (2007) Cognitive stopping rules for terminating information search in online tasks. *MIS Quarterly* 31(1): 89–104.

Brynjolfsson E (1996) The contribution of information technology to consumer welfare. *Information Systems Research* 7(3): 281–300.

Buterin V, Conner E, Dudley R, et al. (2019) EIP-1559: Fee market change for ETH 1.0 chain. Retrieved January 12, 2024. Available at: https://eips.ethereum.org/EIPS/eip-1559.

Cachon GP, Girotra K and Netessine S (2020) Interesting, important, and impactful operations management. *Manufacturing & Service Operations Management* 22(1): 214–222.

Callaway B and Sant'Anna PH (2021) Difference-in-differences with multiple time periods. *Journal of Econometrics* 225(2): 200–230.

Cao J, Ho MS, Ma R, et al. (2024) Transition from plan to market: Imperfect regulations in the electricity sector of China. *Journal of Comparative Economics* 52(2): 509–533.

Capponi A, Ólafsson S and Alsabah H (2023) Proof-of-Work cryptocurrencies: Does mining technology undermine decentralization? *Management Science* 69(11): 6455–6481.

Chung S, Kim K, Lee CH, et al. (2023) Interdependence between online peer-to-peer lending and cryptocurrency markets and its effects on financial inclusion. *Production and Operations Management* 32(6): 1939–1957.

Cui Y, Hu M and Liu J (2023) Value and design of traceability-driven blockchains. *Manufacturing & Service Operations Management* 25(3): 1099–1116.

de Chaisemartin C and D'Haultfoeuille X (2020) Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review* 110(9): 2964–2996.

Dong J, Yom-Tov E and Yom-Tov GB (2019) The impact of delay announcements on hospital network coordination and waiting times. *Management Science* 65(5): 1969–1994.

Dong L, Jiang P and Xu F (2023) Impact of traceability technology adoption in food supply chain networks. *Management Science* 69(3): 1518–1535.

Ecken P and Pibernik R (2015) Hit or miss: What leads experts to take advice for long-term judgments? *Management Science* 62(7): 2002–2021.

Fang Z, Ho Y-C, Tan X, et al. (2021) Show me the money: The economic impact of membership-based free shipping programs on E-tailers. *Information Systems Research* 32(4): 1115–1127.

Feldman P, Li J and Tsai H-T (2022) Welfare implications of congestion pricing: Evidence from SF park. *Manufacturing & Service Operations Management* 24(2): 1091–1109.

Ferreira MV, Moroz DJ, Parkes DC, et al. (2021) Dynamic posted-price mechanisms for the blockchain transaction-fee market. In: *Proceedings of the 3rd ACM conference on advances in financial technologies*, pp.86–99. Arlington Virginia.

Gaba V, Lee S, Meyer-Doyle P, et al. (2023) Prior experience of managers and maladaptive responses to performance feedback: Evidence from mutual funds. *Organization Science* 34(2): 894–915.

Goldstein DG, McAfee RP, Suri S, et al. (2020) Learning when to stop searching. *Management Science* 66(3): 1375–1394.

Guo P and Zipkin P (2007) Analysis and comparison of queues with different levels of delay information. *Management Science* 53(6): 962–970.

Han J, Popkowski Leszczyc PT and Zhang Z (2021) Empirical analyses of nonlinear effects of reserve prices on ending prices in online auctions. *Journal of Interactive Marketing* 54(1): 86–102.

Hinz O, Hann I-H and Spann M (2011) Price discrimination in e-commerce? An examination of dynamic pricing in name-your-own price markets. *MIS Quarterly* 35(1): 81–98.

Ilk N, Shang G, Fan S, et al. (2021) Stability of transaction fees in bitcoin: A supply and demand perspective. *MIS Quarterly* 45(2): 563–692.

Johnson DD and Fowler JH (2011) The evolution of overconfidence. *Nature* 477(7364): 317–320.

Kahn-Lang A and Lang K (2020) The promise and pitfalls of differences-in-differences: Reflections on 16 and pregnant and other applications. *Journal of Business & Economic Statistics* 38(3): 613–620.

Kostami V and Ward AR (2009) Managing service systems with an offline waiting option and customer abandonment. *Manufacturing & Service Operations Management* 11(4): 644–656.

Lee YS and Siemsen E (2017) Task decomposition and newsvendor decision making. *Management Science* 63(10): 3226–3245.

Liu Y, Lu Y, Nayak K, et al. (2022) Empirical analysis of eip-1559: Transaction fees, waiting times, and consensus security. In: *Proceedings of the 2022 ACM SIGSAC conference on computer and communications security*, pp.2099–2113. Los Angeles.

Logg JM, Minson JA and Moore DA (2019) Algorithm appreciation: People prefer algorithmic to human judgment. *Organizational Behavior and Human Decision Processes* 151: 90–103.

Malik N, Aseri M, Singh PV, et al. (2022) Why bitcoin will fail to scale? *Management Science* 68(10): 7323–7349.

Mayya R and Huang P (2025) Startup accelerators, information asymmetry, and corporate venture capital investments. *Management Science*. ePub ahead of print Mar 3.

Mithas S, Chen ZL, Saldanha TJV, et al. (2022a) How will artificial intelligence and industry 4.0 emerging technologies transform operations management? *Production and Operations Management* 31(12): 4475–4487.

Mithas S, Xue L, Huang N, et al. (2022b) Editor's comments: Causality meets diversity in information systems research. *MIS Quarterly* 46(3): iii–xviii.

Pagnotta ES (2022) Decentralizing money: Bitcoin prices and blockchain security. *The Review of Financial Studies* 35(2): 866–907.

Papalexopoulos T, Alcorn J, Bertsimas D, et al. (2023) Reshaping national organ allocation policy. *Operations Research* 72(4): 1475–1486.

Park J, Konana P, Gu B, et al. (2013) Information valuation and confirmation bias in virtual communities: Evidence from stock message boards. *Information Systems Research* 24(4): 1050–1067.

Piet J, Fairoze J and Weaver N (2022) Extracting godl [sic] from the salt mines: Ethereum miners extracting value. Working Paper. https://doi.org/10.48550/arXiv.2203.15930.

Pun H, Swaminathan JM and Hou P (2021) Blockchain adoption for combating deceptive counterfeits. *Production and Operations Management* 30(4): 864–882.

Reijsbergen D, Sridhar S, Monnot B, et al. (2021) Transaction fees on a honeymoon: Ethereum's EIP-1559 one month later. In: 2021 *IEEE international conference on blockchain (Blockchain)*, pp.196–204. Melbourne.

Roughgarden T (2020) Transaction fee mechanism design for the Ethereum blockchain: An economic analysis of EIP-1559. Working Paper. https://doi.org/10.48550/arXiv.2012.00854.

Shang G, Ilk N and Fan S (2023) Need for speed, but how much does it cost? Unpacking the fee-speed relationship in bitcoin transactions. *Journal of Operations Management* 69(1): 102–126.

Shen B, Dong C and Minner S (2022) Combating copycats in the supply chain with permissioned blockchain technology. *Production and Operations Management* 31(1): 138–154.

Soll JB, Palley AB and Rader CA (2022) The bad thing about good advice: Understanding when and how advice exacerbates overconfidence. *Management Science* 68(4): 2949–2969.

Sriraman S, Wuttke D, Rosenzweig E, et al. (2023) Blockchain-enabled supply chain financing (BCF). Working Paper. https://ssrn.com/abstract=4653151.

Trautmann ST and Van de Kuilen G (2015) Reserve price and competing bids: Reference points for product evaluations in online auctions? *Journal of Consumer Behaviour* 14(4): 285–296.

Tsoukalas G and Falk BH (2020) Token-weighted crowdsourcing. *Management Science* 66(9): 3843–3859.

Van den Bos W, Li J, Lau T, et al. (2008) The value of victory: Social origins of the winner's curse in common value auctions. *Judgment and Decision Making* 3(7): 483–492.

Wang J and Hu M (2020) Efficient inaccuracy: User-generated information sharing in a queue. *Management Science* 66(10): 4648–4666.

Wang Y, Gou G, Liu C, et al. (2021) Survey of security supervision on blockchain from the perspective of technology. *Journal of Information Security and Applications* 60: 102859.

Wei Y and Dukes A (2020) Cryptocurrency adoption with speculative price bubbles. *Marketing Science* 40(2): 241–260.

Xu J, Benbasat I and Cenfetelli RT (2013) Integrating service quality with system and information quality: An empirical test in the e-service context. *MIS Quarterly* 37(3): 777–794.

Yu Q, Allon G and Bassamboo A (2021) The reference effect of delay announcements: A field experiment. *Management Science* 67(12): 7417–7437.

Zhan Y, Yeung ACL, Tan KH, et al. (2025) Success and failure of blockchain technology providers: founders' power, beyond-blockchain exploration and centralized decision-making. *Journal of Operations Management*, ePub ahead of print Mar 25.

Zhao X, Ai P, Lai F, et al. (2022) Task management in decentralized autonomous organization. *Journal of Operations Management* 68(6-7): 649–674.

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